
Entropy: Measuring Visual Complexity

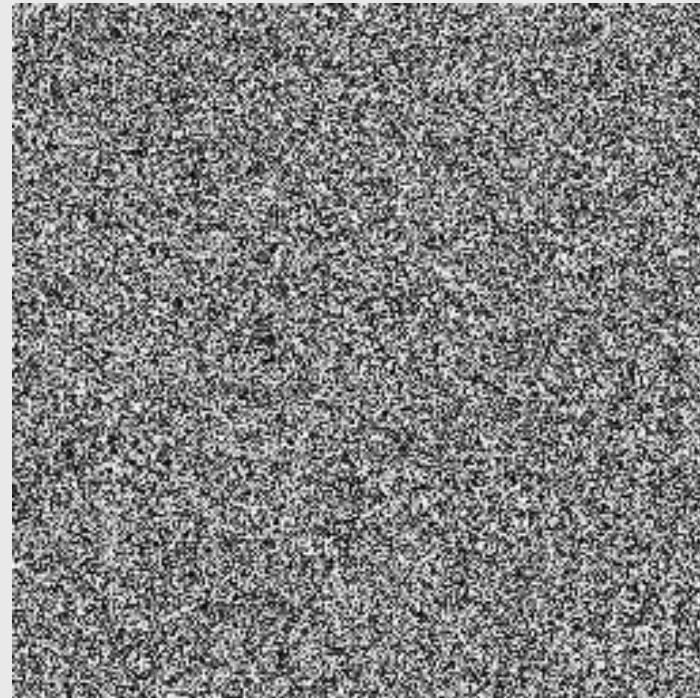
Presented by Junrui Ding

The background of the slide is a light gray with a subtle pattern of small dots. Overlaid on this are several thick, diagonal stripes in various colors including orange, blue, green, purple, and yellow. These stripes are not perfectly straight and have some irregular, organic shapes. Scattered throughout the background are numerous small circles and dots in the same color palette as the stripes, adding to the visual complexity.

Entropy—measures disorder



Completely predictable. Low entropy



Impossible to predict. High entropy.

Shannon Entropy Formula

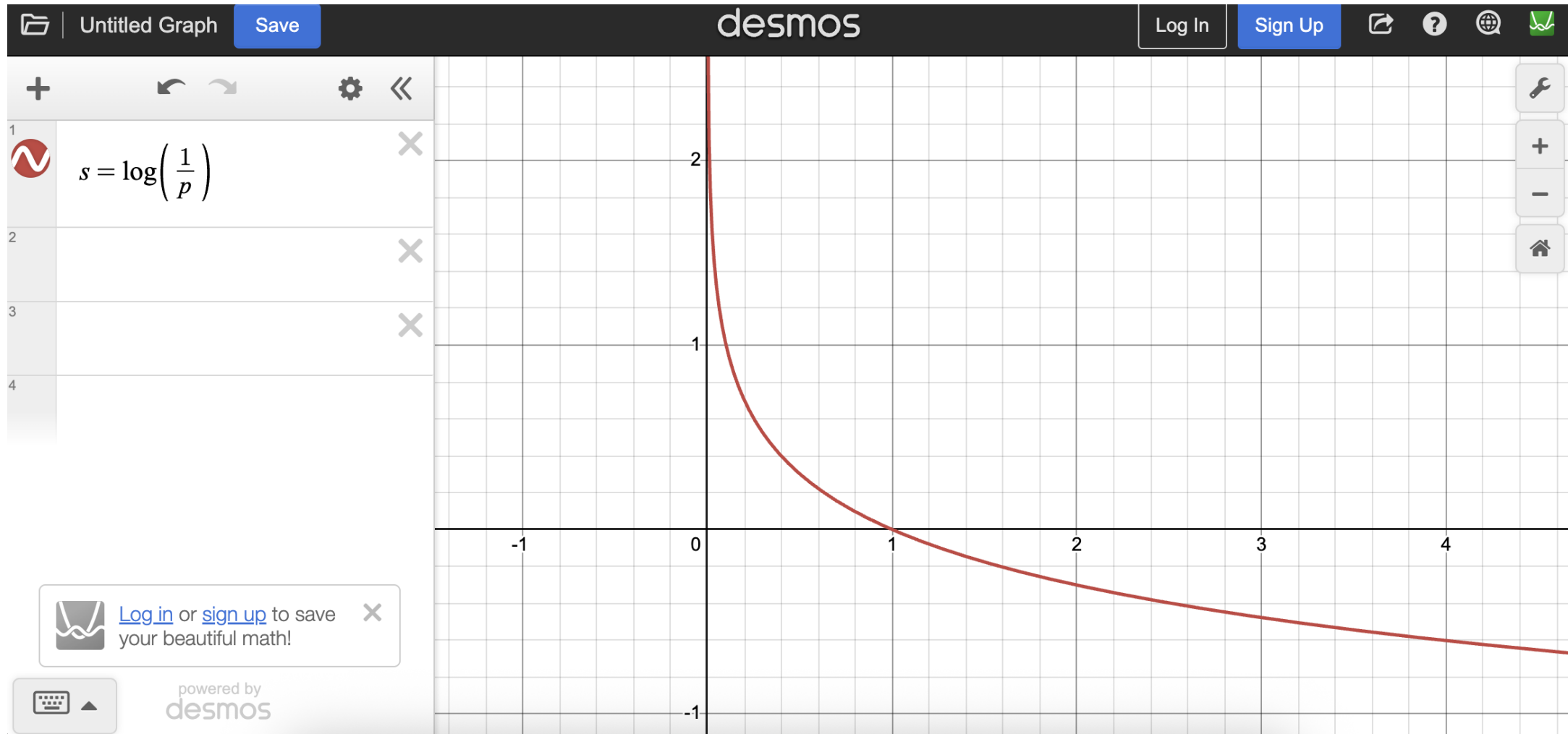
- $H = - \sum_i p_i \log_2 p_i$

p_i = probability of pixel value i appearing in the image, we can get it from normalized Histogram

$-\log_2 p_i = \log_2\left(\frac{1}{p_i}\right)$ = how "surprising" it is to see value i

H = weighted average surprise

Shannon Entropy Formula



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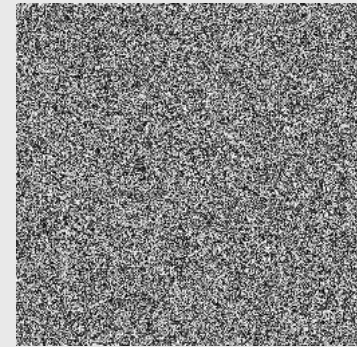
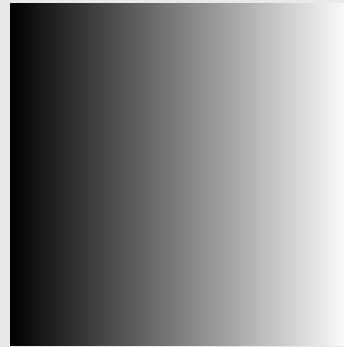
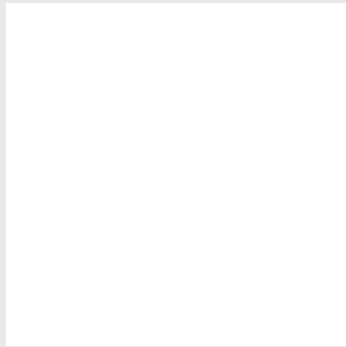
H = weighted average surprise

Computing Image Entropy

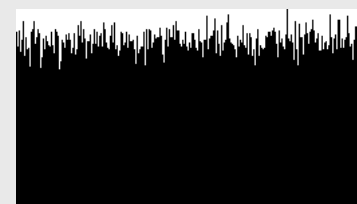
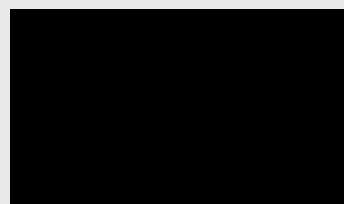
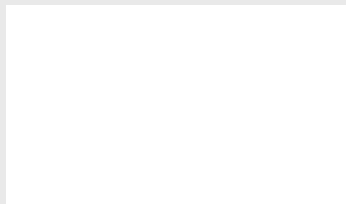
- **Step 1:** Convert image to grayscale
- **Step 2:** Compute normalized histogram → probability distribution
- **Step 3:** Plug into Shannon's formula

Computing Image Entropy

image



histogram



entropy

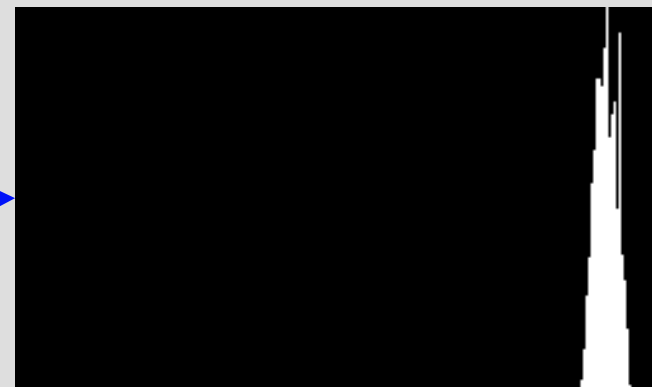
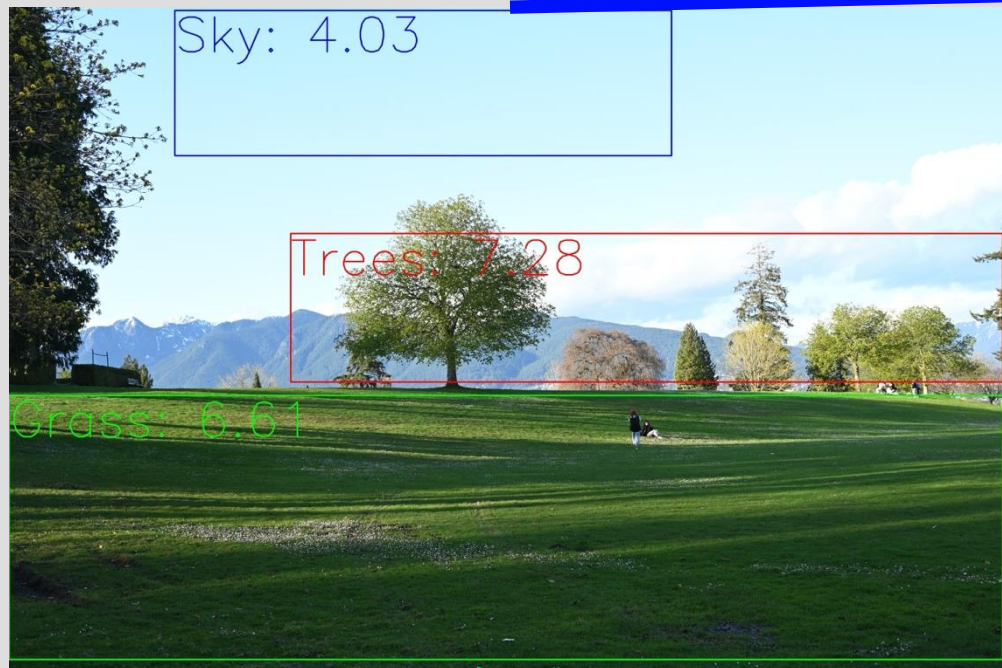
White entropy: 0

Gradient entropy: 8

Noise entropy: 7.99703

Entropy is spatially blind

Entropy as a Feature



Low entropy →
uniform
regions (sky)



High entropy →
rich texture
regions (grass)



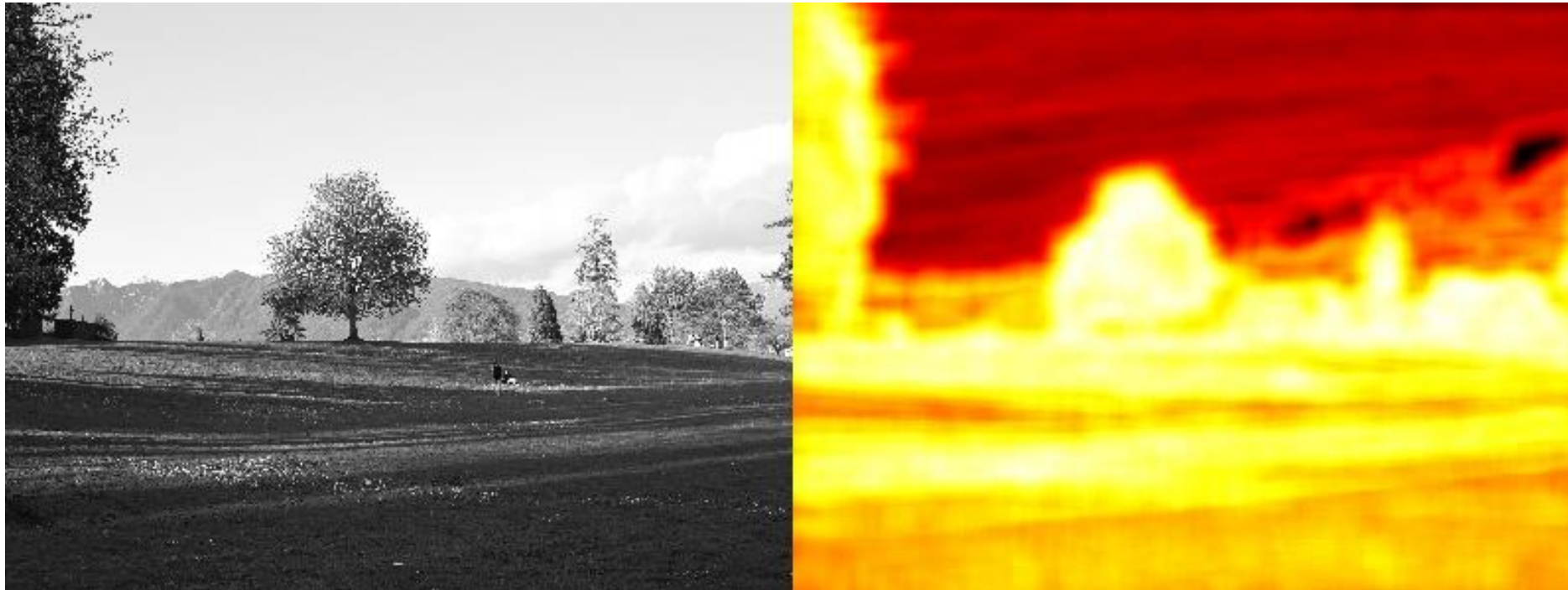
Local Entropy Map

COLORMAP_HOT

Python: cv.COLORMAP_HOT



hot



Limitations and Summary

- Entropy = average unpredictability of pixel values in a region
Computed from the normalized histogram
- High entropy → complex texture; Low entropy → uniform region
- Limitation: spatially blind — structure is invisible to entropy, but can be addressed by computing entropy locally